AI-Enhanced Battery Management Systems for Electric Vehicles: Advancing Safety, Performance, and Longevity

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Abstract. Electric vehicles (EVs) are essential to lowering carbon emissions and solving global environmental issues. The battery powers EVs, making its management crucial to safety and performance. As a self-check system, a Battery Management System (BMS) ensures operating dependability and eliminates catastrophic failures. As batteries age, internal resistance increases and capacity decreases, hence a BMS monitors battery health and performance in real time. EV energy storage systems (ESSs) need a complex BMS algorithm to maintain efficiency. Using battery efficiency calculations that account for charging time, current, and capacity, this approach should reliably forecast the battery's SoC and SoH. As batteries age, internal resistance increases, reducing constant current (CC) charging time. By analyzing these changes, the SoH can be predicted more precisely. Conventional methods for estimating SoC and enhancing BMS performance, such as deep neural networks, are used to minimize error rates. However, as the battery ages, AI approaches have gained prominence for their ability to provide precise diagnostics, fault analysis, and thermal management. These AI-driven techniques significantly enhance safety and reliability during charging and discharging cycles. To further ensure safety, a fault diagnosis algorithm is integrated into the BMS. This algorithm proactively addresses potential issues, thus maintaining the efficiency and safety of the battery. The effectiveness of the proposed BMS algorithms are demonstrated through its successful application in an ESS, validating its capability to manage the battery's state, enhance performance, and ensure operational sustainability in EVs.

1 Introduction

BMSs are essential for the safe and efficient functioning of battery-powered equipment including EVs, renewable energy storage systems, and portable gadgets. They monitor and manage battery characteristics including voltage, current, temperature, and state of charge (SOC) to avoid overcharging, deep draining, and thermal runaway. Proper management of these parameters is essential to optimize battery performance and ensure safety [1-3]. The primary functions of a BMS include state estimation, cell balancing, and fault detection, all of which rely on advanced algorithms and technologies. State estimation, a BMS key function, offers crucial battery SOC, SOH, and SOP information [4]. Recently developed deep learning methods like CNNs and LSTM networks increase accuracy and flexibility. These methods excel in predicting SOC and SOH by modelling complex, non-linear relationships and capturing temporal dynamics.

Cell balancing is another critical function of BMS, ensuring an even distribution of charge among the cells in a battery pack. This process is typically divided into passive and active methods. Passive balancing, which dissipates excess charge as heat, has benefited from advancements in thermal management [5]. On the other hand, active balancing, which involves transferring charge between cells, is being enhanced by developments in power electronics and energy-efficient designs, such as high-efficiency inductors and capacitors that improve energy redistribution. Fault detection and management are essential for identifying and addressing potential issues within the battery system [6]. The use of machine learning algorithms in fault detection has shown superior capabilities compared to traditional model-based approaches. Techniques such as ensemble learning and hybrid models, which combine statistical methods with machine learning, have proven effective in increasing detection accuracy and reliability. Additionally, real-time monitoring systems are being developed to provide immediate feedback and proactive fault management [7-9].

Recent trends in BMS research focus on integrating machine learning and artificial intelligence to enhance state estimation, fault detection, and predictive maintenance. These technologies improve the adaptability of BMS to various operational conditions and enhance decision-making processes. However, challenges remain, particularly in thermal management, algorithm efficiency, and system reliability [10]. Managing thermal performance is critical for

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maintaining battery safety and performance, with recent research exploring advanced cooling techniques and materials to improve heat dissipation. Innovations such as phase change materials and advanced cooling systems are being investigated for better thermal regulation [11]. Improving algorithm efficiency is also a significant challenge, with ongoing efforts to optimize algorithms to reduce computational costs and processing times. This is vital for the practical implementation of advanced techniques in real-time applications. Ensuring system reliability under diverse operating conditions is equally important, with research focused on enhancing system robustness and resilience, particularly in extreme environmental conditions and unexpected operational anomalies [12]. By examining advancements in state estimation, cell balancing, and fault detection, the review aims to provide insights into the current state of the art in battery management systems. Different BMS designs are tailored to handle various battery chemistries and configurations, each offering specific features and capabilities.

2 Objective of the Study

The present study examines battery management systems (BMS) for electric vehicles (EVs) and its integration of sophisticated algorithms for state estimation, fault detection, and predictive maintenance. The project will assess BMS's ability to optimise battery performance, safety, and lifespan using machine learning, artificial intelligence, and temperature management. The research examines existing state-of-the-art technology and approaches to identify obstacles and provide solutions to improve BMS reliability and efficiency under varied operating scenarios.

Gaps in the Literature :

1. **Algorithm Efficiency**: While there has been significant progress in the development of BMS algorithms, there is still a lack of optimization in terms of computational efficiency and real-time application, particularly in complex, non-linear systems.
2. **Thermal Management**: Current studies have not fully addressed the challenges associated with thermal management in BMS, especially in extreme environmental conditions. Advanced cooling techniques and materials are still under exploration, with no definitive solutions widely adopted.
3. **System Reliability**: Although advances in predictive maintenance and fault detection have been made, ensuring system reliability under diverse operating conditions remains a challenge. There is a need for more robust and resilient systems that can operate effectively in unforeseen anomalies and extreme environments.
4. **Data-Driven Approaches**: Despite the growing use of machine learning and AI in BMS, the precision and reliability of these methods depend heavily on the quality and quantity of data available. There is a gap in the availability of comprehensive datasets that can improve the accuracy of BMS predictions and diagnostics.
5. **Integration of Emerging Technologies**: The integration of IoT, cloud computing, and big data in BMS is still in its nascent stage, with many security, privacy, and interoperability concerns yet to be resolved. Further research is needed to address these challenges and fully leverage these technologies for smarter BMS solutions.

3 Battery Management Systems

BMS manages critical functions such as cell balancing, thermal regulation, and state-of-charge monitoring, which are crucial for preserving battery health and safety [13-14]. With the integration of advanced algorithms and sensors, a BMS can precisely oversee and control each cell within the battery pack, thereby preventing overcharging, deep discharging, and overheating issues that can significantly diminish battery lifespan and efficiency. Furthermore, contemporary BMS designs often include predictive analytics and fault detection features, enhancing reliability and enabling proactive maintenance as shown in fig.1. As battery technology advances, BMS also becomes more sophisticated, striving to balance performance, safety, and durability while catering to the specific needs of various applications.



Fig.1. The dashboard of the industrial BMS solution

3.1 Types of batteries

Batteries convert chemical energy into electricity, but the specific materials and technologies can differ. There are five main types of batteries used in today's electric vehicles.

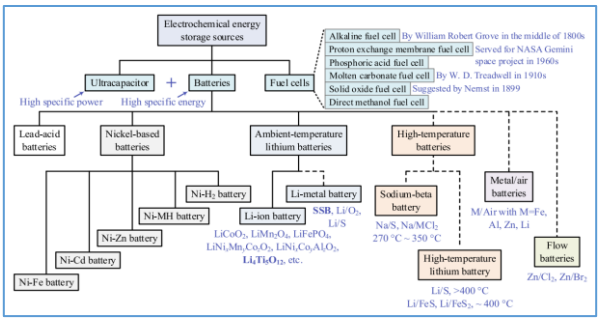


Fig.2.Types of electrochemical energy storage sources.

3.1.1. Lithium-Ion Batteries

Modern electric cars, computers, and smartphones use lithium-ion battery packs due to their excellent

performance, power-to-weight ratio, and high-temperature tolerance. Despite their extensive usage, the development of these batteries has been criticized for its environmental effect, underlining the need for more sustainable alternatives.

3.1.2. Nickel-Metal Hydride Batteries

Nickel-metal hydride batteries are commonly used in hybrid vehicles, despite their higher production costs and lower performance compared to lithium-ion batteries. They offer greater longevity and are more adept at handling frequent charging and discharging, which is typical in hybrids. Additionally, the battery packs in hybrid vehicles are generally smaller than those found in fully electric cars.

3.1.3. Lead-Acid Batteries

Lead-acid battery technology is well-established, known for its affordability and reliability, but it is primarily used as a starter battery in internal combustion engine vehicles. While it is occasionally used in modern electric vehicles, it is limited to auxiliary power systems rather than powering electric motors. Compared to newer battery technologies, lead-acid batteries have a shorter lifespan.

3.1.4. Ultra Capacitors

Ultra capacitors store energy by holding polarized liquid between an electrode and an electrolyte. They are not designed to serve as the main power source but rather as a supplementary battery pack that helps balance the load of the primary lithium-ion battery. Essentially, they act as an intermediary between the main battery and the electric motor, often used to boost acceleration.

3.1.5. Solid-State Batteries

Solid-state batteries are poised to be a major advancement in the electric vehicle sector, with widespread adoption expected in the coming years. Unlike other battery types that use liquid electrolytes, solid-state batteries utilize ceramic materials, making them more environmentally friendly. They also promise greater stability, lower production costs, and easier manufacturing. Experts forecast that solid-state technology could cut battery production costs by up to 40%, representing a significant development.

Table.1. Comparison of Battery power density

Battery Type	Energy Density (Wh/kg)	Life cycle	Toxicity
Li-ion	126-190	500-1,000	Low
Ni-cd	45-80	1,000	High
Ni-MH	100	300-500	Low
Li-ion polymer	185	300-500	Low
Lead acid	30-50	200-300	High
Lithium-Sulphur	55	50-100	Non-toxic

3.2 Battery related issues

Battery Management System (BMS) faces several challenges that can affect the overall performance of a vehicle as shown in fig.3, One issue is battery degradation, which occurs as batteries go through repeated charge and discharge cycles, leading to a gradual loss of capacity and diminished range and efficiency of the electric vehicle [15]. Overcharging is another critical concern; when a battery is charged beyond its maximum voltage, it can overheat and potentially trigger thermal runaway. To mitigate this risk, the BMS must carefully manage charging rates and enforce cutoff limits. Additionally, deep discharging allowing the battery to drop below its recommended level can cause permanent damage, impairing the battery's ability to retain charge and shortening its lifespan [16]. To effectively manage battery-related issues, a sophisticated Battery Management System (BMS) is essential. This system needs to offer real-time monitoring, adaptive algorithms, and comprehensive management strategies. By incorporating advanced sensors, predictive analytics, and robust safety features, the BMS can enhance battery performance, ensure safety, and prolong the lifespan of electric vehicle batteries.

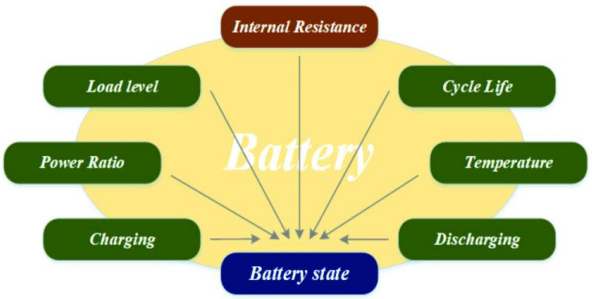


Fig.3. Factors affecting the battery [15]

Key challenges include thermal management, which is vital for maintaining battery efficiency. Excessive heat can accelerate degradation and lead to thermal runaway, while extremely low temperatures can impair battery performance. The BMS must include temperature sensors and manage cooling or heating systems to regulate battery temperature effectively. Cell imbalance is another issue where individual cells within a battery pack charge or discharge at different rates [17]. The BMS needs to implement cell balancing techniques to ensure that all cells charge uniformly, thus promoting consistent performance and longevity.

Voltage fluctuations can stress battery systems and impact performance. The BMS must monitor and control voltage levels to avoid significant deviations that could cause inefficiencies or damage. Additionally, the charging method affects battery health. Fast charging can generate excess heat and shorten battery life, whereas slow charging is less stressful but may be less practical. The BMS must optimize charging strategies to balance efficiency and battery health.
Fault detection and diagnostics are crucial for early issue identification [18]. The BMS should use advanced algorithms to detect real-time anomalies, such as unusual temperature increases or voltage drops, and

issue alerts for maintenance or corrective action. As batteries age, their internal resistance increases, and capacity diminishes, potentially affecting the BMS's accuracy. The BMS must adjust its algorithms to account for these changes and maintain accurate data and predictions. Environmental factors, including humidity, dust, and extreme temperatures, can also impact battery performance and longevity. The BMS should incorporate protective measures to shield the battery from these adverse conditions [19].

3.3 Balancing the Cells in a Battery Pack

Due to variations in state of charge (SOC), state of health (SOH), state of energy (SOE), and state of power (SOP), cell balancing guarantees consistency in charging levels, voltages, and other properties among battery pack cells. Discrepancies in these characteristics might reduce battery performance and capacity. To achieve balance, cells with lower charge need to have their charge, current, or voltage increased, while cells with higher charge require these parameters to be reduced [20]. This balancing process is handled at the hardware level through a charger and monitoring unit, but software is essential for overseeing and controlling the procedure. Balancing resistors in the battery management system (BMS) circuit help facilitate cell balancing by dissipating excess energy from overcharged cells. During charging, one cell is fully charged first, and then the subsequent cells are charged while preventing overcharging of the initial cell. This method ensures uniform voltage across all cells in the battery pack.

3.4 Imbalance charging

Imbalance in lithium-based battery systems occurs due to the repeated cycles of charging and discharging, leading to fluctuations in the charge levels of individual cells over time. Contributing factors include temperature variations, differences in the chemical composition of cells, and their initial state of charge. A battery is considered fully discharged when any single cell reaches its maximum discharge limit, regardless of the charge levels in the other cells. [21].

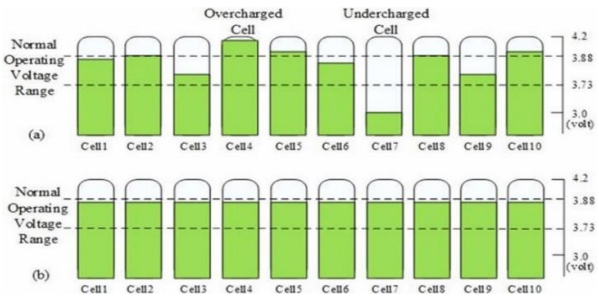


Fig.4. Imbalance vs balanced charging [21]

This can compromise the cell's longevity because it fails to support further discharge. Similarly, during charging, if a specific cell (e.g., cell number 2) becomes fully charged, the entire battery pack must stop charging to prevent damage as shown in fig.4, without protection, overcharging of the other cells can harm the fully charged cell. The primary focus of this research is to explore how the Battery Management System (BMS) addresses these issues by balancing the cells, improving energy efficiency, and extending the overall life of the battery. Electric vehicles (EVs) need dependable battery safety diagnostics and fault-handling methods. Even tiny errors may damage batteries. Battery Management Systems (BMS) have complex sensor, actuator, and battery failure mechanisms. Current, voltage, and temperature sensors may fail [22]. Current sensor failures may reduce the accuracy of states of charge, energy, health, and remaining usable life estimations. Batteries must stay within voltage and temperature restrictions to avoid performance deterioration and safety issues. Sensor faults in voltage and temperature readings can also result in errors in thermal management systems or disrupt battery equalization, as illustrated in Figure 5.

Actuator faults include issues with fuses, high-voltage contactors, the controller area network (CAN) bus, terminal connectors, and cooling systems. Faulty connections can increase resistance, generate excessive heat, and accelerate temperature rises, while poor connections may lead to insufficient power supply and potentially melt battery terminals, creating safety hazards. Inadequate cooling can prevent the battery from staying within safe temperature ranges, risking thermal runaway.

Battery failures include overheating, overcharging, over-discharging, internal and external short circuits, swelling, electrolyte leakage, and thermal runaway. Side reactions from overcharging and over-discharging may cause battery swelling and leaking. Gases generated during thermal runaway contribute to battery swelling and leaking, jeopardizing its safety.

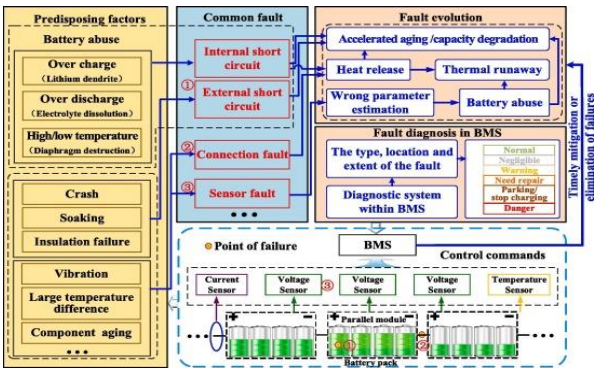


Fig.5. Multi-fault detection and diagnosis of battery pack in EV [22]

3.5 Challenges and foresight

In automotive propulsion, high-energy batteries face significant challenges related to energy density, fast charging, and safety. Addressing these challenges is crucial for advancing carbon neutrality and replacing gasoline vehicles with electric vehicles (EVs). Efficient electric propulsion depends on high energy density to support both rapid acceleration and extended range. Fast charging is also a priority, with goals to charge batteries to 80% capacity in just a few minutes. Additionally, ensuring battery safety is essential for

protecting both people and property, but this should not come at the expense of energy density [23]. To tackle these challenges, future battery technologies are expected to evolve in three main areas: lithium-ion batteries (LIBs), lithium-metal batteries (LMBs), and technologies beyond lithium. LIBs are likely to remain prevalent in EV applications in the near future. For fast-changing needs, lithium titanate-based technologies could offer long cycle life, although they may not be as energy-dense as other options. Lithium-metal anodes are considered promising, but developing practical Li–O2 batteries will require significant research [24]. Alternative ion batteries, such as sodium-ion batteries (SIBs), are also being explored. These batteries show competitive performance and may work alongside LIBs in future EVs. Research is focusing on various chemistries within LIBs, LMBs (like solid-state batteries), and non-lithium technologies (such as sodium and magnesium) to meet the diverse requirements of different applications. These advancements aim to provide batteries with lifetimes of at least 2000 cycles and over 10 years, suitable for EVs [25]. Additionally, the development of sensor-on-chip technology is expected to improve Battery Management Systems (BMS), enhancing their ability to manage and optimize battery performance.

4 Optimisation Techniques in BMS

4.1 State of charge estimation SOC

The ratio of a battery's current accessible capacity to its theoretical capacity is its State of Charge (SOC). SOC is usually stated as a percentage via the equation:

$$Q \text{ remaining SOC} = Q \text{ total} \times 100\%$$

State of Charge (SOC) cannot be directly measured as a physical quantity. However, accurately estimating the current SOC is crucial for ensuring that a battery system delivers its maximum capacity and provides a reliable assessment of the remaining driving range [26]. SOC also helps the Battery Management System (BMS) calculate SOH, SOP, charging currents, and battery balance [26]. State of charge (SOC) estimate accuracy has improved since the 1980s. Onboard battery management systems (BMS) use ampere-hour integral computation, lookup table, and model-based SOC estimate approaches. Figure 6 shows how each of these methods optimizes battery system management and performance.

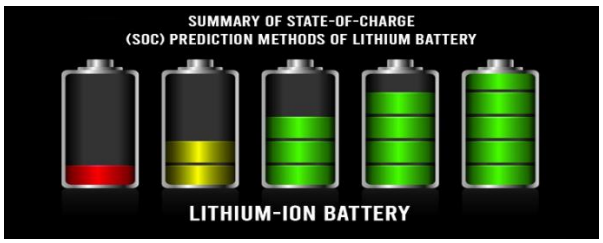


Fig.6. State of charge [26]

4.2 State of health prediction SOH

State of Health (SOH) measures the charge capacity of an aged battery cell relative to that of a new one, providing insight into the battery's ability to meet operational demands. SOH is essential for assessing the current condition of a battery and estimating when it may require replacement. Various techniques for predicting SOH in lithium-ion batteries include feed-forward algorithms, regression and probabilistic models, recurrent neural networks, and entropy-based approaches [27]. A 100% SOH represents a brand-new battery, while the SOH value decreases as the battery ages. Monitoring SOH helps determine the battery's current condition and when it might need to be replaced, as shown in Figure 7.

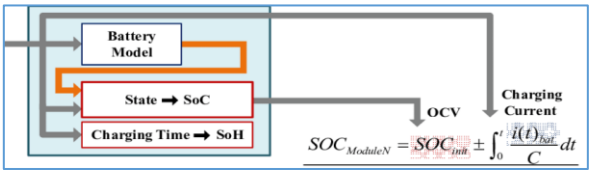


Fig.7. State of health prediction [27]

4.3 State-of-energy (SOE)

SOE Demonstrates how the battery's remaining energy compares to its maximum and usable energy [28]. Figure 8 shows how to estimate this by considering the whole battery charge and discharge cycle.

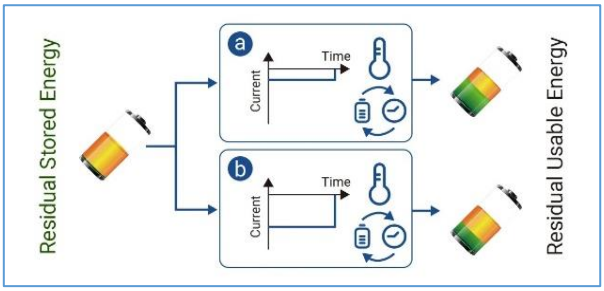


Fig.8 Tracking the battery's remaining available energy and its maximum available energy [28]

4.4 State-of-power (SOP)

As shown in Figure 9, State of Power (SOP) is a battery's maximum power for a certain application. Discharging, operational voltage, and temperature affect SOP.

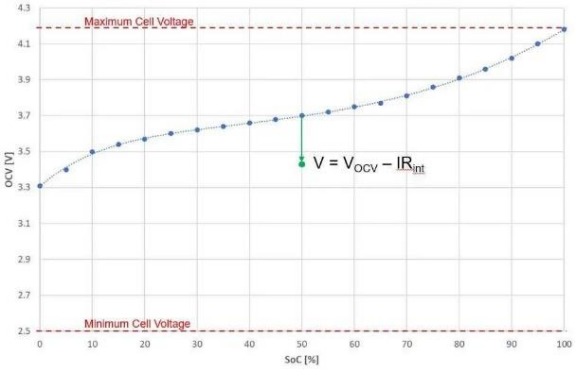


Fig.9. Behaviour of battery under State-of-power observation [29]

SOP is estimated by multiplying operational voltage by maximum discharge current. SOP may vary when the battery degrades and the rated current changes. This approach improves measurements, helping create improved Battery Management Systems. [29].

$$P = U * I$$

4.5 Self-discharge algorithm

Self-discharge occurs when a battery is left in an open circuit and loses capacity. A battery's self-discharge rate (SDR) indicates its capacity retention under different situations. Mechanism classifies self-discharge as irreversible or reversible [30]. As seen in Figure 10, irreversible self-discharge causes capacity loss that cannot be restored in subsequent charging cycles.

In contrast, reversible self-discharge restores capacity over consecutive charging cycles. Reversible self-discharge cells have a larger charge capacity than discharge capacity within a cycle. Metal impurities or burrs in manufacture can cause this. Self-discharge rates may cause SOC discrepancies, which can hasten battery pack capacity deterioration. Traditional techniques for measuring series cell self-discharge rate (SDR) include capacity direct measurement, open-circuit voltage (OCV) monitoring, and voltage floating. The capacity direct measuring technique is simple: charge the battery to a specified SOC, store it for many days, and observe capacity variations.

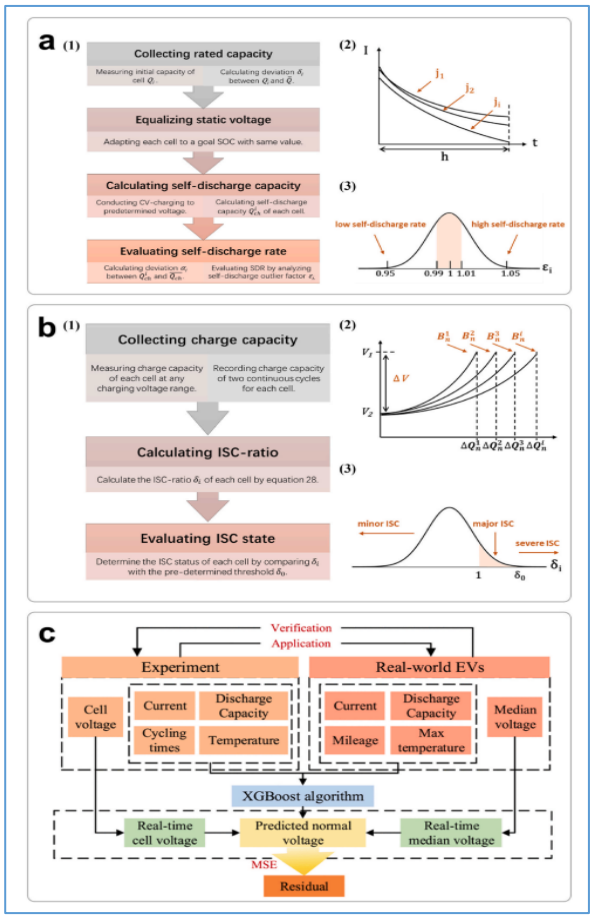


Fig.10 (a) Optimised SDR detection technique schematic (b) Proposed ISC detection method schematic (c) XGBoost-based voltage prediction technique and results.

4.6 Thermal management

Managing battery temperature is crucial for optimizing electric vehicle (EV) performance. High temperatures can trigger thermal runaway in batteries, a condition where exothermic reactions accelerate uncontrollably. When temperatures surpass 90°C, the electrolyte, cathode, and solid electrolyte interface (SEI) layer start to break down. Among lithium-ion batteries, LiFePO4 is notable for its exceptional thermal stability, as it releases less exothermic heat compared to other battery types. Research indicates that for every 1°C rise in temperature within the 30°C to 40°C range, the battery's lifespan can be reduced by about two months.

A well-designed thermal management system (TMS) is essential to maintaining battery temperatures within optimal limits, as depicted in Figure 11. The TMS regulates battery temperature, ensuring that each cell operates within a specified threshold. If the temperature exceeds safe limits, the TMS activates heating or cooling mechanisms to protect the battery and ensure safe operation. This proactive management helps prevent overheating and potential damage.

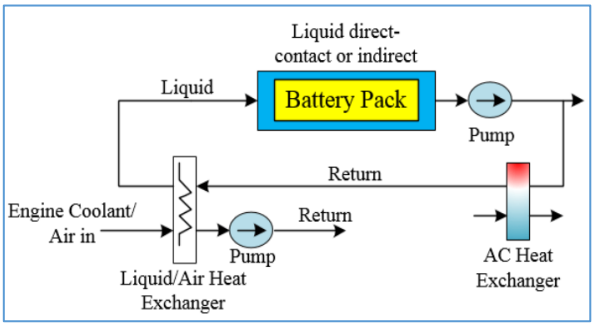


Fig11. Schematic of active thermal management system

4.7 Remaining useful life prediction

Estimating the remaining useful life (RUL) of a battery cell involves predicting the number of load cycles or the duration it will continue to perform effectively, as shown in Figure 12. RUL is a vital indicator that aids in real-time decision-making processes, such as detecting and diagnosing faults.

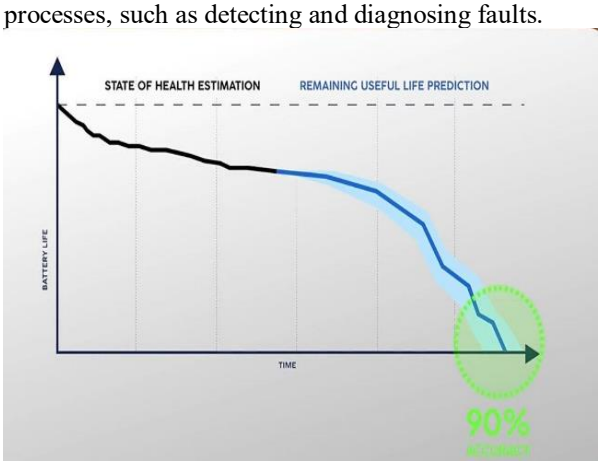


Fig.12. Accuracy Remaining useful life prediction [32]

However, accurately forecasting RUL can be difficult due to numerous uncertainties that may affect the predictions. Various methods exist for estimating RUL,

which can be grouped into several categories: machine learning techniques, time-series analyses, statistical data-driven approaches, Monte Carlo simulations, and additional methodologies [32].

5 Integration of IoT and Cloud Computing Technologies for BMS

In Building Management Systems (BMS), cloud storage, cloud computing, and big data platforms improve AI algorithm and controller performance and reliability. Figure 13 shows how big data technologies enable enhanced analytical methods and vast storage, processing, and analysis. With these technologies, important battery characteristics including SOC, SOH, RUL, thermal runaway, and defect detection can be monitored and saved in the cloud throughout the battery's lifespan [33]. After pre-processing, the battery monitoring and control center will evaluate the data to provide insights and suggestions for future improvements. However, connecting the Internet of Things (IoT) with cloud computing raises security, privacy, interoperability, scalability, data management, standards and protocols, and legal and regulatory difficulties. Therefore, substantial study is needed to address these issues.



Fig.13. Integration of IoT and Cloud Computing [33]

The Internet of Things (IoT) allows for effective data transmission to the cloud, leading to the development of digital twins for battery systems. This digital representation enables the monitoring of the battery's charge levels and aging processes through diagnostic algorithms. The performance and dependability of cloud-based battery management systems have been validated through practical field tests in both stationary and mobile contexts. Although many studies have suggested energy scheduling strategies that utilize cloud computing, these methods frequently face obstacles such as heavy reliance on CPU disk space, constraints of local storage, and specific network setup requirements [34]. Alternative: Power Generation Control and Energy Management (PGCEM) employs a static, multi-metric machine learning approach for effective energy scheduling. Figures 14,15,16 illustrate that the Intelligent Cloud Computing for Power Generation Control and Energy Management (ICC-PGCEM) system incorporates energy generation monitoring, energy storage systems, energy usage sensors, and energy consumption calculators [35]. Most data processing occurs in the cloud, and IoT sensors communicate using different protocols and technologies, with the selection of communication methods influenced by several key factors.

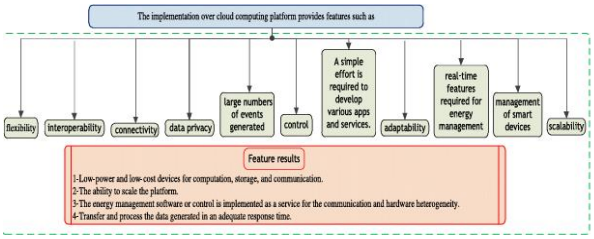


Fig.14. Battery-powered cloud computing [34]

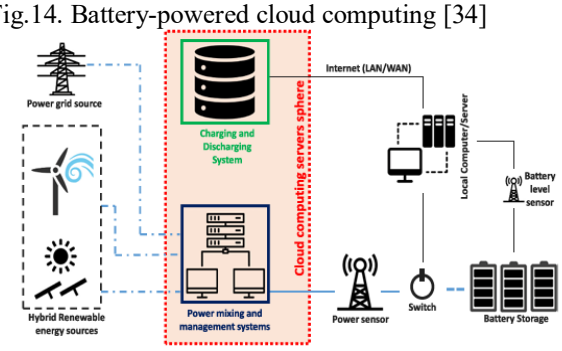


Fig.15. Cloud computing framework to extend battery life

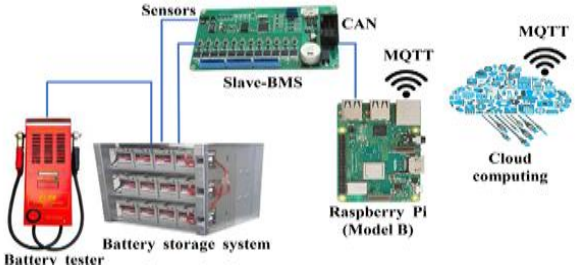


Fig.16 Unified Li-ion and lead-acid battery testing bench with cloud-based BMS prototype for SOC, DoD, and SOH algorithms [35]

6 Machine Learning Techniques for BMS

Artificial Neural Networks (ANNs) were created to simulate human brain information processing.

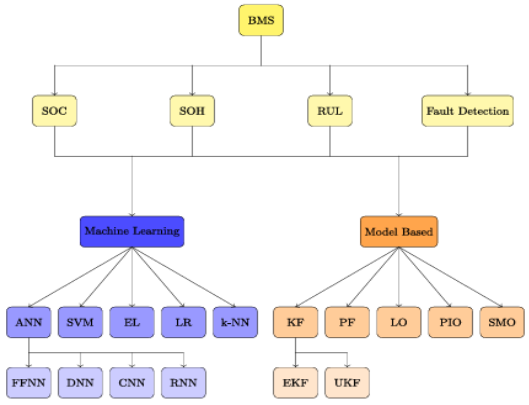


Fig.17. Different BMS algorithm [36]

An ANN has an input layer, one or more hidden layers, and an output layer. The following stages explain ANN operation:

- The input layer receives data, which is sent to the first hidden layer.
- Each neuron in the hidden layer calculates a weighted sum, applies an activation function,

and transmits the outcome to the next hidden layer.

- The process continues through multiple hidden layers until the output layer generates the final prediction or result.

This design allows ANNs to perform well in diverse applications, making them valuable tools for tackling complicated problems [36]. Figure 17 shows how ANNs' computational operations are divided into neural networks and deep learning algorithms by structure and depth. Effective battery management requires accurate SOC measurement. Traditional SOC estimation techniques like coulomb counting are inaccurate because to measurement mistakes, age, and temperature fluctuations. In contrast, Neural Networks (NNs) can capture complicated correlations between battery characteristics and SOC, providing more reliable and precise SOC predictions even with noise and uncertainty. Neural networks may be trained on substantial battery performance and degradation datasets to forecast battery longevity, which is crucial for battery system safety and dependability. Traditional optimization methods use basic heuristics and may miss the complicated relationships between battery properties and charging or discharging situations. Convolutional Neural Networks (CNNs), known for image processing, include many hidden units in their convolutional layers but fewer in their fully linked layers. Recurrent Neural Networks (RNNs) for sequence prediction may need variable numbers of hidden units depending on input sequence complexity and length.

6.1 Feed-forward Neural Network

As shown in Figure 18, the Feedforward Neural Network (FFNN) is a basic artificial neural network that flows information from input to output nodes via any hidden nodes. This design ensures unidirectional data flow by eliminating cycles and feedback loops in node connections [37-38]. The number of layers, the number of neurons per layer, and the activation functions used all affect the network's performance and efficacy.

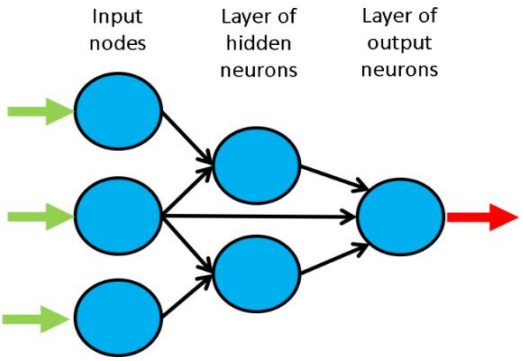


Fig.18. basic Feed-forward Neural Network [37]

Due to its capacity to simulate complicated input-output interactions, Feedforward Neural Networks (FFNNs) are utilized in classification, prediction, and feature extraction. A downside of FFNNs is their sluggish learning rate. Due of the sluggish gradient-based learning methods used to train neural networks and the requirement to repeatedly alter all network parameters, this problem arises. FFNNs are used to analyze battery voltage while charging. This reduces error margins and computing needs. FFNNs can estimate the State of Health (SOH) with upper and lower limits using discharge voltage entropy, incorporating local voltage fluctuations and battery deterioration trends [39]. Monotonic input feature restrictions may enhance FFNN model generalization. Advanced FFNN designs may outperform standard methods. Integrating FFNNs with battery models allows combined estimation of SOC and SOH, which reduces training time and helps find local minima. This method has worked well in low-power embedded systems where FFNNs anticipate Li-Ion battery cell SOC in electric cars.

6.2 Deep Neural Network

Artificial Neural Networks (ANNs) are used in Deep Learning (DL) to detect complex data characteristics. This approach is particularly useful for reliably assessing battery State of Health (SOH) across varying starting SOC and C-rate currents, as shown in Figure 19. SOH is separated into five periods from 100% to 80% [40].

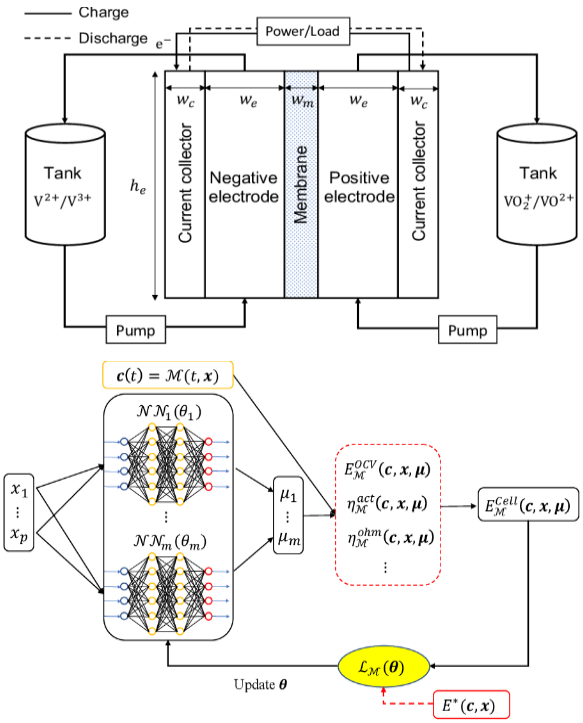


Fig.19. Constrained deep neural network method [40]

In battery management, the first-order derivative of the discharge curve is often used for estimate. Deep Neural Networks (DNNs) optimize estimates. Various DNN models and learning algorithms have been created to improve State of Charge (SOC) estimates and battery management techniques for lithium-ion batteries.

The models used for SOC estimation in electric vehicles feature varying numbers of hidden layers. It has been observed that increasing the number of layers generally reduces the error rate, leading to more accurate SOC predictions [41-42]. This enhancement is especially beneficial when training data is collected from lithium-ion batteries exposed to drive cycle loads under varying ambient temperatures, which boosts

estimation accuracy across different temperature conditions. Newer versions avoid lithium-ion battery overcharging and over discharging, resulting in quicker convergence and more accurate State of Charge (SOC) assessment [43]. Additionally, non-electrical characteristics have been investigated for SOC estimation. This method analyses capacity degradation during lengthy charge/discharge cycles and uses Deep Neural Networks (DNNs) to predict RUL based on varied charge and discharge parameters.

6.3 Convolutional Neural Network

Battery Management Systems are increasingly using CNNs for state predictions. This network has convolutional, pooling, and fully linked layers in each hidden layer. Figure 20 shows that CNNs are multilayer perceptron's since each neuron is linked to the next. CNNs estimate LIB SOH using voltage, current, and charging capacity measurements from partial charging cycles [44-45]. CNNs reduce memory use, improve prediction accuracy, and speed up testing. With STM32Cube.AI, a pre-trained CNN model may be implemented on a microcontroller for real-time SOH estimate. This approach works well for uncertain aging conditions and difficult operating settings.

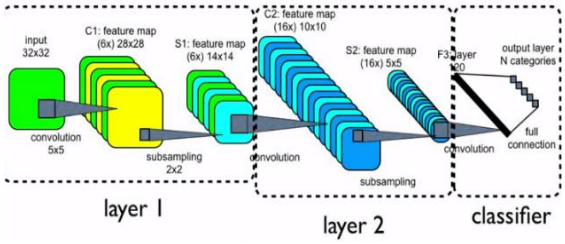


Fig.20. Convolutional Neural Network [44]

Recent works have used one-dimensional CNN algorithms to investigate how noise affects estimate skills and presented a transfer learning methodology for electric cars. CNNs are extensively used in RUL estimation, a crucial part of predictive maintenance across sectors. These models optimise hyper parameters and network performance. By identifying and categorizing damaged Lithium-Ion Battery sensor and communication data, a CNN-driven technique may increase BESS security and dependability. CNN models can also detect bad battery balance circuits with 95.31% accuracy and a high F1 score.

6.4 Recurrent Neural Network

Recurrent Neural Networks (RNNs) are a type of neural network architecture that extends the capabilities of Feedforward Neural Networks (FFNNs) as depicted in figure 21. They are particularly effective for processing sequential data due to their ability to maintain temporal context. However, RNNs can struggle with long-term dependencies in sequences, a challenge known as the long-term dependency problem [46]. To improve model performance, techniques such as sparse sampling can be employed, which helps in managing the variety and complexity of input signals.

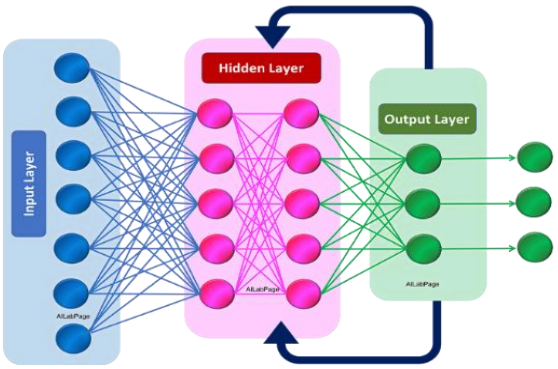


Fig.21. Recurrent Neural Network [46]

Battery characteristics can be effectively represented using compact data, and optimization techniques that exploit signal sparsity enable the recovery of signals with minimal data while retaining crucial information. To maximize the benefits of the Long Short-Term Memory (LSTM) method, improve the network design for easier online deployments [47]. When used in lithium-ion battery modeling and State of Charge (SOC) estimation with time-delayed RNNs, Recurrent Neural Networks (RNNs) can capture complex non-linear battery behaviours and address gradient issues, helping the network learn long-term capacity degradation relationships. RNNs can also estimate aero-engine RUL, enabling health monitoring.

6.5 Machine Learning Algorithms

With the broad deployment of sensing equipment and the fast evolution of IoT devices, data collecting has become easier, allowing for a more detailed digital depiction of batteries. Machine learning, support vector machines, radial basis functions, and recurrent neural networks are gaining popularity [48].

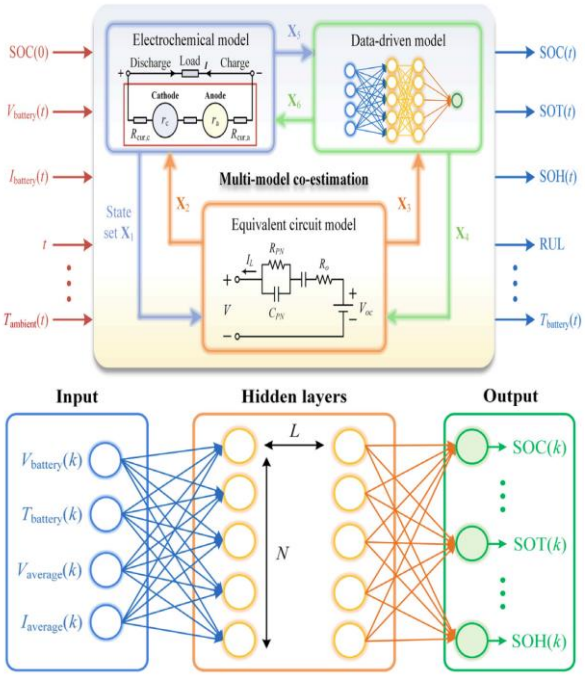


Fig.22. Estimating battery status using neural network [48]

Integrating AI into Battery Management Systems (BMS) offers a promising opportunity to create a

battery-information twin, which provides a sophisticated framework for more accurate and efficient battery state estimation as shown in fig.22. Research has shown successful applications of these technologies, such as artificial neural network models for predicting lithium-ion battery temperature and the use of genetic algorithms and improved optimization techniques for managing battery state of charge. Additionally, deep neural network structures have proven effective in battery state estimation. These AI-based tools are expected to significantly enhance future BMSs [49]. However, emerging fields will present multidisciplinary challenges and require advancements in data security, information communication, and other relevant areas as shown in Fig.23.

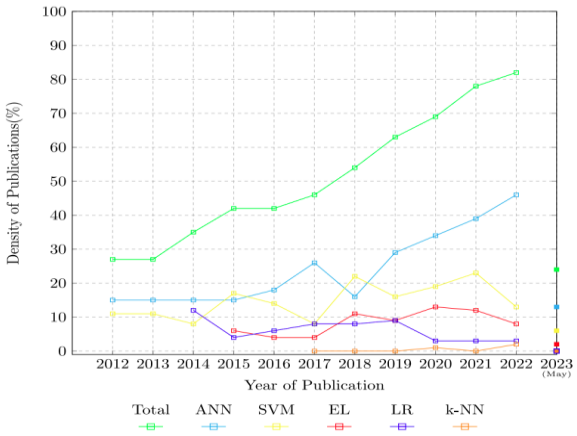


Fig.23. Statistics of WoS indexed publications on Machine Learning based BMS [49]

7 BMS Issues and Challenges

Implementation, battery, and algorithmic issues are the main problems.

7.1 Algorithms issues

Intelligent approaches may help create successful Battery Management Systems (BMS), however they have limits. FNNs can perform well, but storage and execution speed limits limit their usefulness. Noise and uncertainty can be managed using regression and probabilistic approaches, but high-dimensional and non-linear systems are difficult. Deep learning techniques accurately predict SOC, SOH, and RUL, but they need big, high-quality datasets and plenty of processing power. Optimization algorithms improve performance when used with machine learning and deep learning, but they may confront local minima, restricted search capabilities, and unsuitable parameter settings [50]. The entropy algorithm is useful for quantifying signal complexity but requires significant computational resources. The Monte Carlo (MC) method is effective in addressing probabilities and complex distributions, yet it demands substantial computational power. Statistical data-driven approaches can perform well in noisy environments and under degradation, but they often experience slow convergence due to intensive mathematical calculations. Fuzzy controllers offer reliability and adaptability but require extensive data and expertise to function effectively. Model Predictive Control (MPC) enhances the transient response and manages multiple variables but is challenged by its high computational burden and complex algorithms.

7.2 Implementation issues

Implementing a Battery Management System (BMS) may be difficult and impact its performance and dependability. Accurate cell state of charge (SoC) and health evaluation is crucial. Accurate sensors and algorithms are needed to monitor voltage, current, temperature, and battery longevity. This method is complicated by cell production inconsistencies, cell characteristics, and environmental conditions. To avoid overcharging and deep draining, which may damage battery performance and safety, cells must be balanced. Batteries work well within particular temperature ranges; therefore, a well-designed thermal management system is necessary to minimize overheating or cooling [51]. The BMS must interact with other systems in the vehicle or device to avoid operational faults or safety risks. Adjustments to integrate the BMS with existing hardware and software might increase development time and expense. Compliance with safety standards and regulations requires extensive testing and validation to satisfy industry requirements. Solving these issues is crucial to improving battery efficiency, safety, and lifespan for electric cars and other battery-powered technology [52].

7.3 Data abundance and data variety

Data volume and diversity are major challenges in Battery Management Systems (BMS). Sensors that measure each cell's voltage, current, and temperature create a lot of data in modern battery packs. To make real-time analysis and operational choices from this much data, advanced processing and storage solutions are needed. Additionally, raw sensor outputs, derived metrics like state of charge and health, and operating circumstances must be smoothly linked and evaluated. Diversity adds complexity since the BMS must correlate data types, such as temperature data with voltage and current measurements to handle thermal management and prevent overheating. For accurate battery life estimates and efficient charging techniques, the system must adapt to ambient variables and use patterns using sophisticated algorithms and machine learning [53]. System design and integration are further complicated by ensuring data from multiple sources is compatible with various hardware and software components. The BMS needs efficient data handling for cell balance, fault detection, and safety monitoring. Advanced data analytics and strong infrastructure are needed to interpret and use the vast and diverse data to improve battery performance, dependability, and longevity.

7.4 Overall issues

Battery Management Systems (BMS) state estimate accuracy is affected by battery chemistry, aging cycles, thermal runaway dangers, capacity degradation, and charging and balancing issues. Variations in battery chemistry may affect state estimation accuracy even with comparable intelligence algorithms. Lithium Iron Phosphate (LiFP) batteries are cheap and safe, whereas

Lithium Titanate (LiTO) batteries last long. The recurrent Nonlinear Auto-Regressive Exogenous (NARX) method has root mean square errors (RMSE) of 0.53% for LiFP batteries and 0.7% for LiTO batteries, demonstrating battery type-specific performance differences. Aging cycles change internal resistance, capacitance, anode and cathode structural integrity, and solid electrolyte interphase (SEI) layer thickness, complicating state determination. State-of-charge, state-of-health, and remaining usable life forecasts might differ because to this decline. Generalized Particle Filter (GPM) RUL prediction tests found RMSE values of 1.58% and 1.3% for 60 and 80 cycles, respectively. Thermal runaway is another major issue for intelligent algorithms and control techniques. As SOC increases and age progresses, mechanical or electrical stress, lithium plating, overcharging, internal short circuits, exothermic reactions, and impacts increase the likelihood of this occurrence. Models have been developed to limit the dangers of the negative electrode, SEI layer, and electrolyte breaking down over 90°C. BMS algorithm performance is also affected by capacity and power degradation. Discharge circumstances degrade the battery's active material, reducing capacity and power. Increased resistance and lower operating voltage decrease power. Temperatures exceeding 45°C and aging cycles accelerate deterioration. More capacity loss might result from high voltage. Optimization of heat management, efficiency, lifespan, and health depends on charging processes. Rapid charging may overheat EVs, whereas slow charging may limit acceptance. Charge imbalances across batteries due to physical qualities, deterioration, material faults, manufacturing inconsistencies, and tolerances might affect BMS state estimates and safety. Effective charge balance improves clever algorithm performance and ensures safe and dependable battery management.

8 Conclusions

To improve and accept electric cars, sophisticated Battery Management Systems (BMS) must be developed and integrated. As EV batteries become more complicated and demanding, advanced BMS that can precisely predict SOC, SOH, and RUL are needed. Under changing operating circumstances, AI and machine learning may improve these estimations' accuracy and dependability, according to the report. The BMS's strong fault detection algorithms identify possible faults early, boosting battery system safety and lifespan. Cooling methods and materials must improve to maintain battery performance and safety. Cloud computing and IoT boost BMS capability with real-time monitoring and predictive maintenance. The paper concludes that BMS technologies must be researched and developed to meet the EV industry's growing needs, manage modern battery technologies, and contribute to sustainable transportation.

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